

Content

Learning objectives and content

Objectives

After following this course, students

1. understand basic principles of software engineering and identify trends in the automotive software development
2. can describe modern automotive E/E architectures
3. can apply the equations of motion used in vehicle dynamics, that govern the motions of a road vehicle when considered as a point mass.
4. interpret basics of road loads, drive cycles, and powertrain topologies used in powertrain system design analysis

Content

The course consists of four modules of equal duration, and of equal weight in the course grade. Modules 1 and 2 are examined together with an individual project. Modules 3 and 4 are examined in a written closed-book exam.

1. Module 1 Automotive Software
2. Module 2 Electrical/Electronic (E/E) Automotive Architectures
3. Module 3 Vehicle Dynamics
4. Module 4 Electric and Hybrid Powertrains

2026

Spectrum of automotive (5ATB0)

Practical information

Course module

5ATB0

Course type

Bachelor College

Credits (ECTS)

5 EC

Category

Introductory

Language of instruction

English

Follow-up subjects

5AIC0 (Vehicle Networking) and 4AUB20 (Road Vehicle Dynamics)

Offered by

Electrical Engineering - Electronic Systems

Course enrolment

OSIRIS Student

You can register yourself via OSIRIS Student. To do so, go to Enroll / Course.

Enrolment periods

- Block 3

Timeslot(s)

D - We 5-8, Tu 9-10, Fr 1-4

Enrolment period

15 November 2026 until 3 January 2027 23:59

Unenrolment

until 1 February 2027 23:59

Entrance requirements

Assumed previous knowledge

Programming in C, preferably basic Matlab knowledge

Previous knowledge can be gained by

Programming and Engineering Challenge - 5EWCO

Instructional modes

Instructional modes

- Instruction with notebook

Remark

Explanation

Combination of lectures and instruction

Tests

Tests

- Practical / Experiment

Interim examination

Test weight

50

Minimum grade

- Written examination

Final examination

Test weight

50

Minimum grade

5.0

Test enrolment

OSIRIS Student

You can register yourself via OSIRIS Student. To do so, go to Enroll / Test.

Practical / Experiment

- Block 3
Enrolment period
15 November 2026 until 21 March 2027 23:59

Written examination

- Block 3
Enrolment period
15 November 2026 until 21 March 2027 23:59

Unenrolment
until 21 March 2027

Test date
8 April 2027

- Block 4
Enrolment period
15 November 2026 until 6 June 2027 23:59

Unenrolment
until 6 June 2027

Test date
1 July 2027

Lecturer(s)

Co-lecturer

- dr. M. Nabi , MSc
- dr.ir. I. Barosan
- dr.ir. I.J.M. Besselink
- prof.dr.ir. T. Hofman

Contactperson for the course

- prof.dr. K.G.W. Goossens

Responsible lecturer

- prof.dr. K.G.W. Goossens

Subject matter expert

- prof.dr. K.G.W. Goossens

Materials

Required material

- PYNQ board and its display board

Recommended material

- Yanja Dajsuren, Mark van den Brand (eds), Automotive systems and software engineering. Available online <https://link.springer.com/book/10.1007/978-3-030-12157-0>

- J.L. Meriam & L.G. Kraige

Engineering Mechanics Dynamics

8th edition

John Wiley & Sons, Inc.

- Lino Guzzella and Antonio Sciarretta

Vehicle Propulsion Systems: Introduction to Modeling and Optimization

Also available only <https://link.springer.com/book/10.1007/978-3-642-35913-2>

ISBN

978-3-642-35912-5

Do you want to know more?

prof.dr. K.G.W. Goossens can tell you more about this course.

Send an e-mail to k.g.w.goossens@tue.nl